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SPRING AI

GENERATIVE ARTIFICIAL INTELLIGENCE CON JAVA

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Corso avanzato per Venis S.p.A, Venezia, Italia

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Note

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for handwriting practice or general writing. There are no margins, text, or other markings on the paper.



IMAGE-TO-TEXT

Note

[illegible]

- Simone Scannapieco

Note

[illegible]

- ## Utilizzo GPT-4.0 per visione computazionale

- ➔ Modo alternativo per implementare ITT attraverso GPT...
- ➔ ...adattabile anche per STT
- ➔ **Media** astrae dalla tipologia di entità multimediale
- ⚠️ Sviluppatore responsabile della scelta di un modello multimodale compatibile con la *task*

Note

[illegible]

➔ Servizio *image-to-text* Gemini

- 1 Creazione interfaccia ed implementazione del servizio
- 2 Modifica controllore MVC
- 3 *Test delle funzionalità con Postman/Insomnia*

Note

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```
package it.venis.ai.spring.demo.services;

import org.springframework.core.io.Resource;

import it.venis.ai.spring.demo.model.Answer;

public interface MultiModalityService {

    Answer getTranscriptionFromAudioFile(Resource audioFile);

    Answer getDescriptionFromImage(Resource imageFile);

}
```

[illegible]

```
package it.venis.ai.spring.demo.services;

import javax.print.attribute.standard.Media;

...

@Service
public class MultiModalityServiceImpl implements MultiModalityService {

    private final ChatClient geminiChatClient;

    public MultiModalityServiceImpl(@Qualifier("geminiChatClient") ChatClient geminiChatClient) {

        this.geminiChatClient = geminiChatClient;
    }
    ...
    @Override
    public Answer getDescriptionFromImage(Resource imageFile) {

        String contentType = getContentType(imageFile);

        Media imageMedia = new Media(MimeTypeUtils.parseMimeType(contentType), imageFile);

        return new Answer(this.geminiChatClient.prompt()
            .user(u -> u.text("Descrivi in lingua italiana la seguente immagine: ").media(imageMedia))
            .call()
            .content());
    }
}
```

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```
...
private String getContentType(Resource multimediaFile) {
    String contentType = URLConnection.guessContentTypeFromName(multimediaFile.getFilename());

    if (contentType == null || contentType.equals("application/octet-stream")) {
        String fileName = multimediaFile.getFilename();
        if (fileName != null) {
            String extension = fileName.substring(fileName.lastIndexOf('.') + 1).toLowerCase();
            contentType = switch (extension) {
                case "mp3" -> "audio/mpeg";
                case "wav" -> "audio/wav";
                case "m4a" -> "audio/mp4";
                case "ogg" -> "audio/ogg";
                case "flac" -> "audio/flac";
                case "jpg" -> "image/jpeg";
                case "jpeg" -> "image/jpeg";
                case "png" -> "image/png";
                case "raw" -> "image/raw";
                default -> "media/unknown";
            };
        } else {
            contentType = "media/unknown";
        }
    }

    return contentType;
}
}
```

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```
package it.venis.ai.spring.demo.controllers;

...

@RestController
@RequestMapping("/gemini/multi-modality")
public class GeminiMultimodalityController {

    private final MultiModalityService multiModalityService;

    public GeminiMultimodalityController(MultiModalityService multiModalityService) {

        this.multiModalityService = multiModalityService;
    }

    ...

    /**
     * Endpoint per l'analisi e descrizione di immagini (Image-to-Text).
     * Carica un'immagine predefinita dal classpath e utilizza le capacità di visione
     * di Gemini per generare una descrizione testuale del contenuto dell'immagine.
     *
     * @param audioFile risorsa immagine dal classpath (file PNG predefinito)
     * @return Answer oggetto contenente la descrizione generata dell'immagine
     */
    @PostMapping("/itt")
    public Answer getDescriptionFromImage(@Value("classpath:multimodal.test.png") Resource audioFile) {

        return this.multiModalityService.getDescriptionFromImage(audioFile);
    }

}
```

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Branch: 17-spring-ai-gemini-ollama-multimodality-itt

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